```
entry:
%0 = icmp eq i32 %num_args, 2, !dbg !15
 br i1 %0, label %assert_end, label %assert_fail, !dbg !15, !prof !16
T | F

assert_end:
%.not = icmp eq ptr %args, null, !dbg !15
 br i1 %.not, label %assert_fail1, label %assert_end2, !dbg !15, !prof !20
T | F

assert_end2:
%var_A.type_index = load i32, ptr %args, align 4, !dbg !15
 %var_A.type_index.fr = freeze i32 %var_A.type_index, !dbg !15
 %3 = icmp sgt i32 %var_A.type_index.fr, 63, !dbg !15
 br i1 %3, label %assert_end4, label %switch.early.test, !dbg !15
T | F

switch.early.test:
switch i32 %var_A.type_index.fr, label %assert_fail3 [
 i32 7, label %assert_end4
 i32 4, label %assert_end4
 i32 0, label %assert_end4
 ], !dbg !15
def | 7 | 4 | 0

assert_fail:
%1 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
... !17
 tail call void %1(ptr nonnull @.str.1, ptr nonnull @.str), !dbg !15
 br label %common.ret, !dbg !15

assert_end4:
%5 = getelementptr inbounds i8, ptr %args, i64 16, !dbg !15
 %var_T_squeeze.type_index = load i32, ptr %5, align 4, !dbg !15
 %var_T_squeeze.type_index.fr = freeze i32 %var_T_squeeze.type_index, !dbg !15
 %6 = icmp sgt i32 %var_T_squeeze.type_index.fr, 63, !dbg !15
 br i1 %6, label %assert_end6, label %switch.early.test43, !dbg !15
T | F

assert_fail1:
%2 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
... !17
 tail call void %2(ptr nonnull @.str.1, ptr nonnull @.str.2), !dbg !15
 br label %common.ret, !dbg !15

switch.early.test43:
switch i32 %var_T_squeeze.type_index.fr, label %assert_fail5 [
 i32 7, label %assert_end6
 i32 4, label %assert_end6
 i32 0, label %assert_end6
 ], !dbg !15
def | 7 | 4 | 0

assert_end6:
%8 = getelementptr inbounds i8, ptr %args, i64 8, !dbg !15
 %9 = load ptr, ptr %8, align 8, !dbg !15
 %10 = icmp eq i32 %var_A.type_index.fr, 72, !dbg !15
 %var_A.idx = select i1 %10, i64 16, i64 0, !dbg !15
 %var_A = getelementptr inbounds i8, ptr %9, i64 %var_A.idx, !dbg !15
 %11 = getelementptr inbounds i8, ptr %args, i64 24, !dbg !15
 %12 = load ptr, ptr %11, align 8, !dbg !15
 %13 = icmp eq i32 %var_T_squeeze.type_index.fr, 72, !dbg !15
 %var_T_squeeze.idx = select i1 %13, i64 16, i64 0, !dbg !15
 %var_T_squeeze = getelementptr inbounds i8, ptr %12, i64 %var_T_squeeze.idx,
... !dbg !15
 %.not44 = icmp eq ptr %9, null, !dbg !15
 br i1 %.not44, label %assert_fail7, label %assert_end8, !dbg !15, !prof !20
T | F

assert_fail3:
%4 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
... !17
 tail call void %4(ptr nonnull @.str.1, ptr nonnull @.str.3), !dbg !15
 br label %common.ret, !dbg !15

assert_fail7:
%14 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %14(ptr nonnull @.str.1, ptr nonnull @.str.5), !dbg !15
 br label %common.ret, !dbg !15

assert_end8:
%15 = getelementptr inbounds i8, ptr %var_A, i64 16, !dbg !15
 %16 = load i32, ptr %15, align 4, !dbg !15
 %17 = icmp eq i32 %16, 2, !dbg !15
 br i1 %17, label %assert_end10, label %assert_fail9, !dbg !15, !prof !16
T | F

assert_end10:
%19 = getelementptr inbounds i8, ptr %var_A, i64 24, !dbg !15
 %squeeze.var_A.shape = load ptr, ptr %19, align 8, !dbg !15
 %20 = load i64, ptr %squeeze.var_A.shape, align 8, !dbg !15, !tbaa !28
 %M = trunc i64 %20 to i32, !dbg !15
 %21 = getelementptr inbounds i8, ptr %var_A, i64 32, !dbg !15
 %squeeze.var_A.strides = load ptr, ptr %21, align 8, !dbg !15
 %22 = getelementptr inbounds i8, ptr %var_A, i64 12, !dbg !15
 %dev_id = load i32, ptr %22, align 4, !dbg !15
 %A = load ptr, ptr %var_A, align 8, !dbg !15
 call void @llvm.assume(i1 true) [ "align"(ptr %A, i64 64) ], !dbg !15
 %.not45 = icmp eq ptr %12, null, !dbg !15
 br i1 %.not45, label %assert_fail11, label %assert_end12, !dbg !15, !prof !20
T | F

assert_fail9:
%18 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %18(ptr nonnull @.str.1, ptr nonnull @.str.6), !dbg !15
 br label %common.ret, !dbg !15

assert_fail11:
%23 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %23(ptr nonnull @.str.1, ptr nonnull @.str.7), !dbg !15
 br label %common.ret, !dbg !15

assert_end12:
%24 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 16, !dbg !15
 %25 = load i32, ptr %24, align 4, !dbg !15
 %26 = icmp eq i32 %25, 1, !dbg !15
 br i1 %26, label %assert_end14, label %assert_fail13, !dbg !15, !prof !16
T | F

assert_end14:
%28 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 24, !dbg !15
 %squeeze.var_T_squeeze.shape = load ptr, ptr %28, align 8, !dbg !15
 %29 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 32, !dbg !15
 %squeeze.var_T_squeeze.strides = load ptr, ptr %29, align 8, !dbg !15
 %T_squeeze = load ptr, ptr %var_T_squeeze, align 8, !dbg !15
 call void @llvm.assume(i1 true) [ "align"(ptr %T_squeeze, i64 64) ], !dbg !15
 %30 = getelementptr inbounds i8, ptr %var_A, i64 22, !dbg !15
 %31 = load i16, ptr %30, align 2, !dbg !15
 %32 = icmp eq i16 %31, 1, !dbg !15
 %33 = getelementptr inbounds i8, ptr %var_A, i64 21, !dbg !15
 %34 = load i8, ptr %33, align 1, !dbg !15
 %35 = icmp eq i8 %34, 32, !dbg !15
 %36 = getelementptr inbounds i8, ptr %var_A, i64 20, !dbg !15
 %37 = load i8, ptr %36, align 1, !dbg !15
 %38 = icmp eq i8 %37, 2, !dbg !15
 %39 = and i1 %35, %38, !dbg !15
 %40 = and i1 %32, %39, !dbg !15
 br i1 %40, label %assert_end16, label %assert_fail15, !dbg !15, !prof !16
T | F

assert_fail13:
%27 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %27(ptr nonnull @.str.1, ptr nonnull @.str.8), !dbg !15
 br label %common.ret, !dbg !15

assert_end16:
%42 = getelementptr inbounds i8, ptr %squeeze.var_A.shape, i64 8, !dbg !15
 %43 = load i64, ptr %42, align 8, !dbg !15, !tbaa !47
 %44 = and i64 %43, 4294967295, !dbg !15
 %45 = icmp eq i64 %44, 1, !dbg !15
 br i1 %45, label %assert_end18, label %assert_fail17, !dbg !15, !prof !16
T | F

assert_fail15:
%41 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %41(ptr nonnull @.str.1, ptr nonnull @.str.9), !dbg !15
 br label %common.ret, !dbg !15

assert_end18:
%.not46 = icmp eq ptr %squeeze.var_A.strides, null, !dbg !15
 br i1 %.not46, label %if_end, label %if_then, !dbg !15, !prof !20
T | F

assert_fail17:
%46 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %46(ptr nonnull @.str.1, ptr nonnull @.str.10), !dbg !15
 br label %common.ret, !dbg !15

if_then:
%47 = load i64, ptr %squeeze.var_A.strides, align 8, !dbg !15, !tbaa !49
 %48 = and i64 %47, 4294967295, !dbg !15
 %49 = icmp eq i64 %48, 1, !dbg !15
 %50 = icmp eq i32 %M, 1, !dbg !15
 %51 = or i1 %50, %49, !dbg !15
 br i1 %51, label %if_end, label %assert_fail19, !dbg !15, !prof !16
T | F

if_end:
%52 = getelementptr inbounds i8, ptr %var_A, i64 40, !dbg !15
 %53 = load i64, ptr %52, align 8, !dbg !15
 %54 = icmp eq i64 %53, 0, !dbg !15
 br i1 %54, label %assert_end22, label %assert_fail21, !dbg !15, !prof !16
T | F

assert_fail19:
%55 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %55(ptr nonnull @.str.1, ptr nonnull @.str.11), !dbg !15
 br label %common.ret, !dbg !15

assert_end22:
%57 = getelementptr inbounds i8, ptr %var_A, i64 8, !dbg !15
 %58 = load i32, ptr %57, align 4, !dbg !15
 %59 = icmp eq i32 %58, 1, !dbg !15
 br i1 %59, label %assert_end24, label %assert_fail23, !dbg !15, !prof !16
T | F

assert_fail21:
%56 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %56(ptr nonnull @.str.1, ptr nonnull @.str.12), !dbg !15
 br label %common.ret, !dbg !15

assert_end24:
%61 = icmp ne ptr %A, null, !dbg !15
 %62 = icmp eq i32 %M, 0, !dbg !15
 %63 = or i1 %62, %61, !dbg !15
 br i1 %63, label %assert_end26, label %assert_fail25, !dbg !15, !prof !16
T | F

assert_fail23:
%60 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %60(ptr nonnull @.str.1, ptr nonnull @.str.13), !dbg !15
 br label %common.ret, !dbg !15

assert_end26:
%65 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 22, !dbg !15
 %66 = load i16, ptr %65, align 2, !dbg !15
 %67 = icmp eq i16 %66, 1, !dbg !15
 %68 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 21, !dbg !15
 %69 = load i8, ptr %68, align 1, !dbg !15
 %70 = icmp eq i8 %69, 32, !dbg !15
 %71 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 20, !dbg !15
 %72 = load i8, ptr %71, align 1, !dbg !15
 %73 = icmp eq i8 %72, 2, !dbg !15
 %74 = and i1 %70, %73, !dbg !15
 %75 = and i1 %67, %74, !dbg !15
 br i1 %75, label %assert_end28, label %assert_fail27, !dbg !15, !prof !16
T | F

assert_fail25:
%64 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %64(ptr nonnull @.str.1, ptr nonnull @.str.14), !dbg !15
 br label %common.ret, !dbg !15

assert_end28:
%77 = load i64, ptr %squeeze.var_T_squeeze.shape, align 8, !dbg !15, !tbaa
... !59
 %78 = trunc i64 %77 to i32, !dbg !15
 %79 = icmp eq i32 %M, %78, !dbg !15
 br i1 %79, label %assert_end30, label %assert_fail29, !dbg !15, !prof !16
T | F

assert_fail27:
%76 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %76(ptr nonnull @.str.1, ptr nonnull @.str.15), !dbg !15
 br label %common.ret, !dbg !15

assert_end30:
%.not47 = icmp eq ptr %squeeze.var_T_squeeze.strides, null, !dbg !15
 br i1 %.not47, label %if_end32, label %if_then31, !dbg !15, !prof !20
T | F

assert_fail29:
%80 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %80(ptr nonnull @.str.1, ptr nonnull @.str.16), !dbg !15
 br label %common.ret, !dbg !15

if_then31:
%81 = load i64, ptr %squeeze.var_T_squeeze.strides, align 8, !dbg !15,
... !tbaa !69
 %82 = and i64 %81, 4294967295, !dbg !15
 %83 = icmp eq i64 %82, 1, !dbg !15
 %84 = icmp eq i32 %M, 1, !dbg !15
 %85 = or i1 %84, %83, !dbg !15
 br i1 %85, label %if_end32, label %assert_fail33, !dbg !15, !prof !16
T | F

assert_fail5:
%7 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
... !17
 tail call void %7(ptr nonnull @.str.1, ptr nonnull @.str.4), !dbg !15
 br label %common.ret, !dbg !15

if_end32:
%86 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 40, !dbg !15
 %87 = load i64, ptr %86, align 8, !dbg !15
 %88 = icmp eq i64 %87, 0, !dbg !15
 br i1 %88, label %assert_end36, label %assert_fail35, !dbg !15, !prof !16
T | F

assert_fail33:
%89 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %89(ptr nonnull @.str.1, ptr nonnull @.str.17), !dbg !15
 br label %common.ret, !dbg !15

assert_end36:
%91 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 8, !dbg !15
 %92 = load i32, ptr %91, align 4, !dbg !15
 %93 = icmp eq i32 %92, 1, !dbg !15
 br i1 %93, label %assert_end38, label %assert_fail37, !dbg !15, !prof !16
T | F

assert_fail35:
%90 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %90(ptr nonnull @.str.1, ptr nonnull @.str.18), !dbg !15
 br label %common.ret, !dbg !15

assert_end38:
%95 = getelementptr inbounds i8, ptr %var_T_squeeze, i64 12, !dbg !15
 %96 = load i32, ptr %95, align 4, !dbg !15
 %97 = icmp eq i32 %dev_id, %96, !dbg !15
 br i1 %97, label %assert_end40, label %assert_fail39, !dbg !15, !prof !16
T | F

assert_fail37:
%94 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %94(ptr nonnull @.str.1, ptr nonnull @.str.19), !dbg !15
 br label %common.ret, !dbg !15

assert_end40:
%99 = icmp ne ptr %T_squeeze, null, !dbg !15
 %100 = or i1 %62, %99, !dbg !15
 br i1 %100, label %assert_end42, label %assert_fail41, !dbg !15, !prof !16
T | F

assert_fail39:
%98 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %98(ptr nonnull @.str.1, ptr nonnull @.str.20), !dbg !15
 br label %common.ret, !dbg !15

assert_end42:
tail call fastcc void @squeeze_compute_(i32 %M, ptr %T_squeeze, ptr %A),
... !dbg !15
 br label %common.ret, !dbg !15

assert_fail41:
%101 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
... !tbaa !17
 tail call void %101(ptr nonnull @.str.1, ptr nonnull @.str.21), !dbg !15
 br label %common.ret, !dbg !15

common.ret:
%common.ret.op = phi i32 [ -1, %assert_fail ], [ -1, %assert_fail1 ], [
... -1, %assert_fail3 ], [ -1, %assert_fail5 ], [ -1, %assert_fail7 ], [ -1,
... %assert_fail9 ], [ -1, %assert_fail11 ], [ -1, %assert_fail13 ], [ -1,
... %assert_fail15 ], [ -1, %assert_fail17 ], [ -1, %assert_fail19 ], [ -1,
... %assert_fail21 ], [ -1, %assert_fail23 ], [ -1, %assert_fail25 ], [ -1,
... %assert_fail27 ], [ -1, %assert_fail29 ], [ -1, %assert_fail33 ], [ -1,
... %assert_fail35 ], [ -1, %assert_fail37 ], [ -1, %assert_fail39 ], [ -1,
... %assert_fail41 ], [ 0, %assert_end42 ]
 ret i32 %common.ret.op, !dbg !15
```

CFG for 'squeeze' function